DSCI 369
Lab 8

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Goals: Visualize inner product and cosine similarity.

Non-MATLAB/Python in-lab portion

Go to `https://www.geogebra.org/m/hfsc8dwg`. This computes the inner product and cosine similarity of $\vec{u}$ and $\vec{v}$. Set

$$\vec{u} = \begin{pmatrix} -2 \\ 2 \end{pmatrix}$$

- Make $\vec{v}$ shorter/longer but pointing in the exact same direction as $\vec{u}$. What happens to the inner product and cosine similarity?
- Make $\vec{v}$ shorter/longer but pointing in the exact opposite direction as $\vec{u}$. What happens to the inner product and cosine similarity?
- Slowly rotate $\vec{v}$ around in a circle. What happens to the cosine similarity?

Exercises

1. Go to `DSCI_369_Module_8_Lab_Python` or `DSCI_369_Module_8_Lab_Matlab`.
2. Go to `DSCI_369_Module_8_Lab_Python` or `DSCI_369_Module_8_Lab_Matlab`.
3. Go to `https://www.geogebra.org/m/hfsc8dwg`. Move around both $\vec{u}$ and $\vec{v}$ to see how the inner product and cosine similarity change. For each of the following, either write the coordinates of the vectors or give a screenshot.
   (a) Slide around the vectors until the cosine similarity is (approximately) 0.5. Is the angle between the vectors acute, obtuse, or right?
   (b) Slide around the vectors until the cosine similarity is (approximately) -0.5. Is the angle between the vectors acute, obtuse, or right?
   (c) Slide around the vectors until the cosine similarity is (approximately) 0. Is the angle between the vectors acute, obtuse, or right?